

The Impact of an Aging Farmer Population on Agriculture

HDSI Agri Datathon 2024

Outdoor Boys

October 6th 2024

Lucca DeFulgentis
Boston University
Commonwealth Ave,
Boston, MA 02215
ludef@bu.edu

Shawn Balgobind
Boston University
Commonwealth Ave,
Boston, MA 02215
srb5012@bu.edu

Moses Chen
Boston University
Commonwealth Ave,
Boston, MA 02215
mosesc@bu.edu

Abstract

This study examines the impact of the increasing average age of farmers on agriculture in the United States. Utilizing data from the Census of Agriculture (2002, 2007, 2012, 2017, 2022) and the USDA's National Agricultural Statistics Service, this research analyzes trends in production, age distribution, and sales among farmers. By identifying the implications of an aging farming population, this study aims to discuss the effects on agricultural production and the possible broader implications for the future.

Year	Average Age
2002	55.3
2007	57.1
2012	58.3
2017	57.5
2022	58.1

Table 1: A simple reformatted table of average age of farmers in the U.S

1. Introduction

Agriculture is crucial for the US economic system and ensures that there is enough food for the continuously growing populations. Farmers are the backbone and keep the entire agriculture industry afloat. However, recent trends show that the average age of farmers are increasing. As retirement approaches for the older generation, this study questions the sustainability and effects on agriculture production.

Since 1997 the average age of US farmers has steadily increased from an average of 55.3 to 58.1 years old. This suggests several problems regarding production capacity and number of farmers. Questions arise such as: Can older farmers implement new production techniques and how efficient can remain as they get older. This study investigates the production of younger farmers compared to older ones through national averages and county averages.

The agriculture industry was valued at \$543.1 in 2022 increasing from the \$101.4 billion in 2002. This increase (not accounting for inflation) demonstrates the importance of understanding the effects of an aging population. Through analysis of age distribution and production trends, this study looks into the future sustainability of U.S agriculture.

2. Data and Methods

Production data was analyzed from the Census of Agriculture census from the USDA's National Agricultural Statistics Service (NASS). The data spanned the years 1997, 2003, 2007, 2012, 2017, 2022. Information such as state FIPS codes, county codes, year and agricultural land harvested in acres was taken and reformatted. A key challenge was handling agricultural land harvested that had disclosure values (D) which were undisclosed information about production. To estimate values, state totals (represented by a 999 FIPs code) had known county production values subtracted from it. The resulting number was divided by the count of the unknown values to average the county production of (D) values. Other values included (z) which meant not statistically significant.

For age data, the study utilized data from Census of Agriculture census from the USDA's National Agricultural Statistics Service again. The data spanned 2003, 2007, 2012, 2017, 2022. Information such as state FIPS code, county codes, year, and age were reformatted into a simple dataframe. Age data was then averaged for states by utiliz-

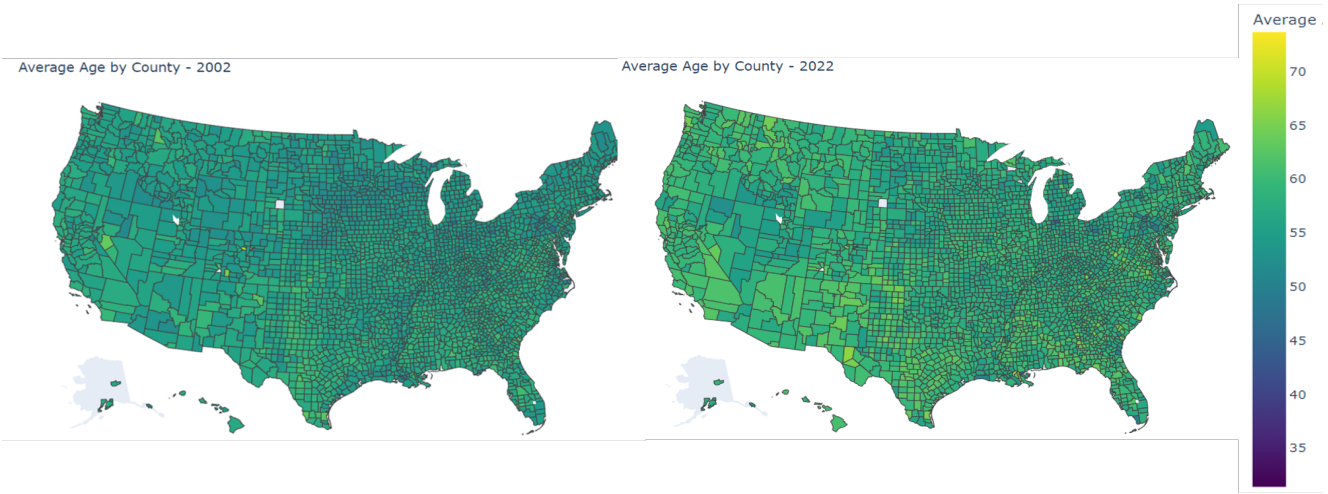


Figure 1: The heat map depicts the average age per county in 2002 and 2022

ing county data. (Table 1)

Both data infrastructures were reformatted through the python library pandas and then graphed through the python library plotly. These datasets included FIPs codes, county codes, year and then their respective data.

A important distinction about the data is after 2017, the census changed the distinction of farmers from operators to producers. Operators would include the farm owner and manager while producers include all farm hands. This skewed the data in 2017 by introducing younger farm hands. This may explain why the average age increases from 2002 to 2012 but decreases in 2017.

The study also formatted the age data by county. Figure 1 depicts 2002 and 2022 and the age averages per county. The average age by county in 2002 depicts an average age of 55.3 nation wide while in 2022 the national average is 58.1. The lighter the green, the older the average age is in the county. From 2002 to 2022, the coloration shows a increase of age. Specifically the west coast where agriculture is traditionally located increased in age.

The study first performed a comparison of averages sales by age group sorted by year (Figure 3 on page 3) This was found by pulling data from the Census once again and creating a new dataframe. This data frame included average income of farms and average age. The study then utilised plotly to create a cluster bar graph showing 5 years 2002, 2007, 2012, 2017, 2022. The y axis was used to shown average age. Each cluster of bars represented sales categories. This plot shows that younger farmers aged make more than their older counterparts.

Lastly, the study performed a person correlation coeffi-

cient. This is the ratio between the covariance of two variables and the product of their standard deviations. The two variables being the average age per county and the acres harvested per county.

$$R = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (1)$$

Coeff: -0.248

1 shows a positive correlation, 0 shows no correlation, -1 will so a negative correlation. In this study we are sufficiently close to zero which shows no correlation between

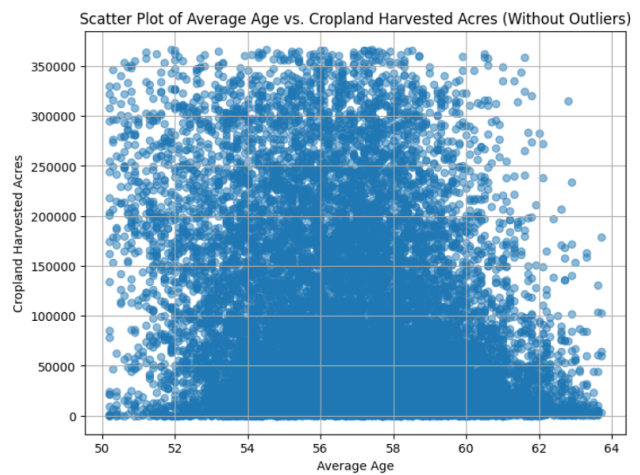


Figure 2: Scatter Plot of Cropland Harvested vs Average Age (outliers removed)

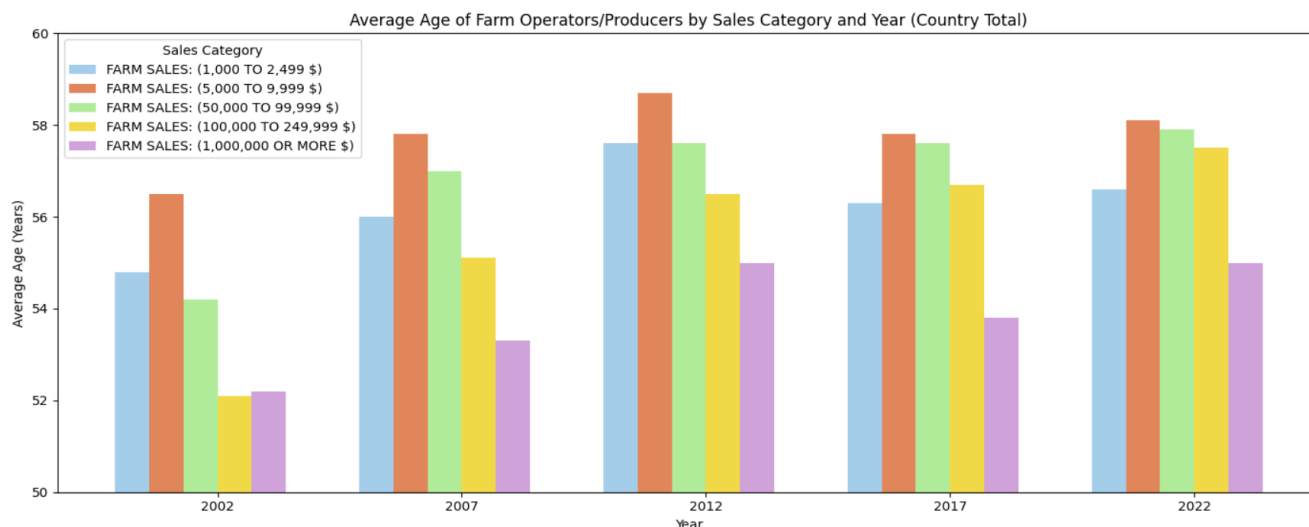


Figure 3: The cluster bar graph depicts the different agriculture sales categories per year and how they relate to age

average age per county and the acres harvested per county.

The study created a scatter plot of all the data. Outliers were calculated by finding the upper and lower bounds of the data. The IQR represents the range within which the central 50% of the data lies. Lower and upper bounds are established based on the IQR. Lower bound: $Q1 - 1.5 * IQR$ Upper bound: $Q3 + 1.5 * IQR$ Any data point falling below the lower bound or above the upper bound is considered a potential outlier.

3. Results

The average age of farmers has trended upwards from 1997 to 2022. In 2002, farmers were approximately 55.3 years old on average, rising to about 58.1 years old by 2022 (table 1). Several factors could contribute to this increasing average, such as fewer younger generations entering the farming industry, longer careers within farming, and the financial investments required to start farming for profit.

The data analysis suggests a clear trend, younger farmers tend to be more productive and yield higher profitability from their farms. Figure 3 partitions levels of sales into groups based on age and clustered in years. The relationship between younger age and higher farm profitability becomes evident through this partitioning. However production as a whole has not decreased.

To account for inflation, the economic impact of farm production was standardized. Inflation, which refers to the rising cost of goods and services over time, erodes purchasing power. For instance, while \$1 could purchase three pieces of corn in 1997, it might only buy one in 2022. Even if production levels remain constant, inflation reduces a farm's economic output. By adjusting for inflation, the comparison of farm impact across years becomes more ac-

curate, allowing for a meaningful evaluation of the economic contribution of farms over time.

As seen in figure 2 its clear that there is no relation between average farmer age and acres harvested. Despite this statistic, its also evident that younger farmers work on more lucrative operations. When taking into account inflation, its clear that the aging farming population will affect the output if the trends continue.

4. Conclusion

The data indicates that while the farmer population is aging, this trend has not yet led to a significant decrease in overall production. However, the rising average age raises concerns about the future sustainability of this trend. If the farming population continues to age the long-term effects on agricultural productivity may become more pronounced. For now, the stable production levels suggest that the current farmer population is managing to offset potential inefficiencies related to age, but a further increase in the average age of farmers may have more noticeable impacts on production in the future.

Please see this link to our [Submission Video](#).

Please see this link to our [Google Colab Notebook](#).

References

[Ser22] USDA National Agricultural Statistics Service. 2022 census of agriculture, 2022. https://www.nass.usda.gov/Publications/AgCensus/2022/Online_Resources/County_Profiles/Oregon/cp410001.pdf.